

THILAK RAJ MURUGAN, MS

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Summary

Backend Software Engineer with **5+ years** of experience building **distributed systems** and **ML infrastructure** at scale.

Technical Skills

Languages: Go, Python, C++, JavaScript, TypeScript, PHP, SQL
Backend: GCP, Docker, Kubernetes, PySpark, HDFS, MySQL, Redis, BigQuery, DataDog, Terraform
AI/ML: PyTorch, LLMs, Numpy, Pandas; **Frontend:** React.js, Redux, Material UI, Jest
Concepts: ML Infrastructure, Microservices, Web Apps, Linux, Caching, CI/CD, APIs, gRPC, NoSQL
Coursework: Distributed Systems, Big Data, Data Structures, Algorithms, ML, Databases, Operating Systems

Experience

American Family Insurance

May 2025 – August 2025

ML Ops Intern - ML Infrastructure Team

Madison, United States

- Built **ML model serving** for batch processing on Google Batch service, scaling periodic inference across **1M+** BigQuery rows per run.
- Developed Infrastructure as Code (IaC) with **Terraform**, GitLab CI/CD, **Docker** and **Python** to provision GCP resources, thus automating the end-to-end deployment pipeline and reducing deployment time by **80%**.

Vimeo

June 2023 – August 2024

Software Engineer 3 - IAM Platform, Creator Experience Team

Bengaluru, India

- Owned a high-throughput **user recommendation service** serving **10M+** users by designing backend APIs using PHP, MySQL, Memcached, Docker, and Datadog, thus increasing free-to-paid user conversion by **20%**.
- Optimized **RBAC** based **authorization systems** by improving MySQL query performance and data access patterns.
- Modernized legacy web pages by rebuilding frontend components with React.js and TypeScript, improving logged-in user experience and boosting user interaction time by **30%**.
- Ideated and built a **video recommendation service** using **Hugging Face's** Sentence transformer based **deep learning** model, **Python** and **Golang** for the Vimeo Jam hackathon.

PropertyGuru Group

July 2021 – May 2023

Software Engineer - Fullstack

Bengaluru, India

- Built and scaled multiple **backend microservices** using **Nest.js**, **Node.js**, and **Symfony (PHP)**, serving **11M+** users across the property-tech industry in South-East Asia.
- Reduced config API latency by **25%** with Redis cache and automated customer cache invalidation, thereby increasing the efficiency of non-engineering teams.
- Built frontend webpages/micro-frontends with **TypeScript**, React.js, Redux and Material UI. Improved code maintainability by writing **unit tests** via Jest.

Hiver

May 2019 – June 2021

Software Engineer

Bengaluru, India

- Owned the overhaul of the **billing** system, boosting revenue growth for the shared inbox product built on Gmail.
- Automated & optimized payment processing using the **Stripe SDK** across **Python** and **PHP** services, leveraging Fast API, webhooks and **RabbitMQ**, and implementing multi-factor authentication (MFA/SCA) for EU transactions.

Projects

Distributed KV Store: Code

- Built a distributed, fault-tolerant **key-value store in Go** with horizontal scalability via sharding, **Raft** based **replication** for strong consistency, and durable storage using bbolt; achieved throughput of **23K ops/sec**.

Scalable Data Infrastructure for Analytics:

- Designed and implemented a scalable **data pipeline** for batch (BigQuery, Dataproc) and stream (**Kafka**, **Spark Streaming**) processing on the NYC Taxi dataset, scanned **80GiB in <10s** using BigQuery's Dremel Engine.

Scaling Distributed Databases:

- Benchmarked TiDB using **Go**, scaled data size to **190 GiB** and analyzed system metrics (90 ms OLTP latency). Load-tested **Cassandra** via NoSQLBench, driving **200M** read/write operations with sub-millisecond latency.

Education

University of Wisconsin - Madison

September 2024 – April 2026

Master's in Data Engineering (Computer Sciences Dept.) - GPA: 3.95

Wisconsin, United States

Indian Institute of Technology Kharagpur

July 2014 – April 2019

Bachelor of Technology (Honors) in Electronics and Electrical Communication Engineering

Kharagpur, India